General Principles & Process of Isolation of Elements

Question1

The purification method based on the following physical transformation is :

$$\underset{(X)}{\text{Solid}} \xrightarrow{\text{Heat}} \underset{(X)}{\text{Vapour}} \xrightarrow{\text{Cool}} \underset{(X)}{\text{Solid}}$$

JEE Main 2025 (Online) 28th January Evening Shift

Options:

A.

Extraction

B.

Crystallization

C.

Distillation

D.

Sublimation

Answer: D

Solution:

The purification method described by the transformation:

$$\underset{(X)}{\text{Solid}} \xrightarrow{\text{Heat}} \underset{(X)}{\text{Vapour}} \xrightarrow{\text{Cool}} \underset{(X)}{\text{Solid}}$$

is known as **Sublimation**.

In sublimation, a solid substance directly transforms into its gaseous phase upon heating and reverts back to the solid phase upon cooling. This technique is commonly used to purify substances, particularly those that sublime more easily than others under the same conditions. Sublimation is effective for separating substances with different volatility, like separating ammonium chloride from a mixture with sodium chloride, as ammonium chloride sublimates while sodium chloride does not.

Question2

Match List I with List II.

	LIST I		LIST II
A.	Reverberatory furnace	I.	Pig Iron
B.	Electrolytic cell	II.	Aluminum
C.	Blast furnace	III.	Silicon
D.	Zone Refining furnace	IV.	Copper

[24-Jan-2023 Shift 1]

Options:

A. A - IV, B - II, C - I, D - III

B. A − I, B − IV, C − II, D − III

C. A - I, B - III, C - II, D - IV

D. A - III, B - IV, C - I, D - II

Answer: A

Solution:

Solution:
Reverberatory furnace: Used for roasting of Copper.
Electrolytic cell : For reactive metal : Al
Blast furnace : Hematite to Pig Iron
Zone Refining furnace: For semiconductors : Si

Question3

The metal which is extracted by oxidation and subsequent reduction from its ore is :
[24-Jan-2023 Shift 2]

Options:

A. Al

B. Ag

C. Cu

D. Fe

Answer: B

Solution:

Solution:
Ag.

$$4\,Ag + 8CN^- + O_2 + 2H_2O \rightarrow 4[Ag(CN)_2]^- + 4OH^-$$

$$2[Ag(CN)_2]^{-1} + Zn \rightarrow 2\,Ag \downarrow + [Zn(CN)_4]^{-2}$$

Question4

Which one of the following reactions does not occur during extraction of copper ?
[25-Jan-2023 Shift 1]

Options:

A. $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$

B. $2\,FeS + 3O_2 \rightarrow 2\,FeO + 2SO_2$

C. $CaO + SiO_2 \rightarrow CaSiO_3$

D. $FeO + SiO_2 \rightarrow FeSiO_3$

Answer: C

Solution:

Solution:

$$CuFeS_2 + O_2 \xrightarrow{\text{Partial roasting}}$$

$$Cu_2S + FeO + SO_2 + \underset{\text{very small}}{FeS} + Cu_2\underset{\text{very small}}{O}$$

$$Cu_2S + O_2 \rightarrow Cu_2O + SO_2$$

$$FeS + O_2 \rightarrow FeO + SO_2$$

$$FeO + SiO_2 \rightarrow FeSiO_3$$

No formation of calcium silicate $(CaSiO_3)$ in extraction of Cu.

Question5

Given below are two statements :-
Statement I :- In froth floatation method a rotating paddle agitates the mixture to drive air out of it.
Statement II :- Iron pyrites are generally avoided for extraction of iron due to environmental reasons.
In the light of the above statements, choose the correct answer from the options given below :-
[25-Jan-2023 Shift 2]

Options:

A. Both Statement I and Statement II are true

B. Statement I is false but Statement II is true

C. Statement I is true but Statement II is false

D. Both Statement I and Statement II are false

Answer: B

Solution:

Solution:
In froth floatation method a rotating paddle draws in air and stirs the pulp.

Question6

The reaction representing the Mond process for metal refining is
[29-Jan-2023 Shift 1]

Options:

A. $Ni + 4\,CO \xrightarrow{\Delta} Ni(CO)_4$

B. $2K[Au(CN)_2] + Zn \xrightarrow{\Delta} K_2[Zn(CN)_4] + 2\,Au$

C. $Zr + 2I_2 \xrightarrow{\Delta} Zr\,I$

D. $ZnO + C \xrightarrow{\Delta} Zn + CO$

Answer: A

Solution:

Solution:
Mond's process uses:
$Ni + 4\,CO \rightarrow [Ni(CO)_4]$

Question7

The major component of which of the following ore is sulphide based mineral?
[29-Jan-2023 Shift 2]

Options:

A. Calamine

B. Siderite

C. Sphalerite

D. Malachite

Answer: C

Solution:

Solution:
Calamine: $ZnCO_3$
Siderite: $FeCO_3$

Sphalerite: ZnS
Malachite: $CuCO_3 \cdot Cu(OH)_2$

Question8

In the extraction of copper, its sulphide ore is heated in a reverberatory furnace after mixing with silica to : [30-Jan-2023 Shift 1]

Options:

A. separate CuO as $CuSiO_3$

B. remove calcium as $CaSiO_3$

C. decrease the temperature needed for roasting of Cu_2S

D. remove FeO as $FeSiO_3$

Answer: D

Solution:

Solution:
The copper ore contains iron, it is mixed with silica before heating in reverberatory furnace. FeO slags off as $FeSiO_3$.
$FeO + SiO_2 \rightarrow FeSiO_3$

Question9

Given below are two statements:
Statement I: During Electrolytic refining, the pure metal is made to act as anode and its impure metallic form is used as cathode.
Statement II: During the Hall-Heroult electrolysis process, purified Al_2O_3 is mixed with Na_3AlF_6 to lower the melting point of the mixture.
In the light of the above statements, choose the most appropriate

answer from the options given below:
[30-Jan-2023 Shift 2]

Options:

A. Statement I is incorrect but Statement II is correct

B. Both Statement I and Statement II are incorrect

C. Statement I is correct but Statement II is incorrect

D. Both Statement I and Statement II are correct

Answer: A

Solution:

Solution:
In Electrolytic refining, the pure metal is used as cathode and impure metal is used as anode.
Na_3AlF_6 is added during electrolysis of Al_2O_3 to lower the melting point and increase conductivity.

Question10

The methods NOT involved in concentration of ore are
(A) Liquation
(B) Leaching
(C) Electrolysis
(D) Hydraulic washing
(E) Froth floatation
Choose the correct answer from the options given below :
[31-Jan-2023 Shift 1]

Options:

A. B, D and C only

B. C, D and E only

C. A and C only

D. B, D and E only

Answer: C

Solution:

Solution:
Methods involved in concentration of one are
(i) Hydraulic Washing
(ii) Froth Flotation
(iii) Magnetic Separation
(iv) Leaching

Question11

Which one of the following statements is incorrect? [31-Jan-2023 Shift 2]

Options:

A. Boron and Indium can be purified by zone refining method.

B. Van- Arkel method is used to purify tungsten.

C. Cast iron is obtained by melting pig iron with scrap iron and coke using hot air blast.

D. The malleable iron is prepared from cast iron by oxidising impurities in a reverberatory furnace.

Answer: B

Solution:

Solution:
Van - Arkel process is used for purification of Ti, Zr, Hf and B.

Question12

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R
Assertion A: In an Ellingham diagram, the oxidation of carbon to carbon monoxide shows a negative slope with respect to temperature.
Reason R: CO tends to get decomposed at higher temperature.
In the light of the above statements, choose the correct answer from

the options given below
[1-Feb-2023 Shift 1]

Options:

A. Both A and R are correct and R is the correct explanation of A

B. A is not correct but R is correct

C. Both A and R are correct but R is NOT the correct explanation of A

D. A is correct but R is not correct

Answer: D

Solution:

Solution:
$2C(s) + O_2(g) \rightarrow 2CO(g)$
$\Delta_r S^0$ is +ve, $\Delta_r G^0 = \Delta_r H^0 - T\,\Delta_r S^0$; thus slope is negative
As temperature increases $\Delta_r G^0$ becomes more negative thus it has lower tendency to get decomposed.

Question13

Among following compounds, the number of those present in copper matte is _____.

A. $CuCO_3$

B. Cu_2S

C. Cu_2O

D. FeO

[1-Feb-2023 Shift 2]

Answer: 1

Solution:

Solution:
FeS and Cu_2S, present in copper matte.

Question14

Which of the following metals can be extracted through alkali leaching technique?
[8-Apr-2023 shift 1]

Options:

A. Cu

B. Au

C. Pb

D. Sn

Answer: D

Solution:

Solution:
Sn due to Amphoteric nature.

Question15

In Hall-Heroult process, the following is used for reducing Al_2O_3 :-
[8-Apr-2023 shift 2]

Options:

A. Magnesium

B. Graphite

C. Na_3AlF_6

D. CaF_2

Answer: B

Solution:

Solution:
In case of Hall's process, reduction of Al_2O_3 to Al can be done using graphite.

Question16

Which of the following is used as a stabilizer during the concentration of sulphide ores?
[10-Apr-2023 shift 1]

Options:

A. Xanthates

B. Fatty acids

C. Pine oils

D. Cresols

Answer: D

Solution:

Solution:
Cresol is used as stabilizer

Question17

In the extraction process of copper, the product obtained after carrying out the reactions
(i) $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$
(ii) $2Cu_2O + Cu_2S \rightarrow 6\,Cu + SO_2$ is called
[11-Apr-2023 shift 1]

Options:

A. Reduced copper

B. Blister copper

C. Copper matte

D. Copper scrap

Answer: B

Solution:

Solution:

$$2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 3SO_2$$

$$2Cu_2O + Cu_2S \rightarrow \underset{\text{Blister copper}}{6\,Cu} + SO_2$$

Due to evolution of SO_2, the solidified copper formed has a blistered look and is referred to as blister copper.

Question18

Given below are two statements :
Statement I : In the metallurgy process, sulphide ore is converted to oxide before reduction.
Statement II : Oxide ores in general are easier to reduce.
In the light of the above statements, choose the most appropriate answer from the options given below:
[11-Apr-2023 shift 2]

Options:

A. Both Statement I and Statement II are correct

B. Statement I is correct but Statement II is incorrect

C. Statement I is incorrect but Statement II is correct

D. Both Statement I and Statement II are incorrect

Answer: A

Solution:

Solution:

$2\,ZnS + 3O_2 \rightarrow 2\,ZnO + 2SO_2$

Oxides on carbon reduction forms CO_2 while sulphide on carbon reduction gives CS_2.

CO_2 is more volatile compared to CS_2 therefore oxides are easy to reduce.

Question19

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R
Assertion A : In the Ellingham diagram, a sharp change in slope of the line is observed for Mg → MgO at ~1120°C
Reason R: There is a large change of entropy associated with the change of state
In the light of the above statements, choose the correct answer from the options given below
[12-Apr-2023 shift 1]

Options:

A. Both A and R are true but R is NOT the correct explanation of A

B. A is false but R is true

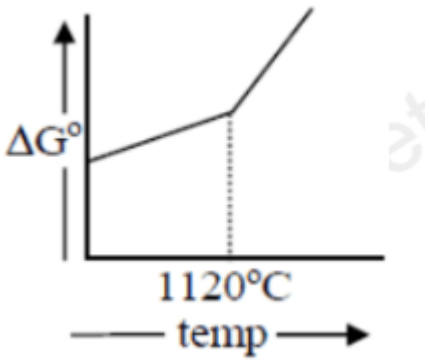
C. A is true but R is false

D. Both A and R are true and R is the correct explanation of A

Answer: D

Solution:

Solution:



(A) shape change slope of $1120^{\circ}C$
(R) During slate change there is large change in entropy.

Question20

Which one of the following is most likely a mismatch? [13-Apr-2023 shift 1]

Options:

A. Zinc - Liquation

B. Copper - Electrolysis

C. Titanium - van Arkel Method

D. Nickel - Mond process

Answer: A

Solution:

Solution:
Zinc is refined by distillation method, which is used for metals having low boiling point.

Question21

Given below are statements related to Ellingham diagram : Statement I : Ellingham diagram can be constructed for oxides, sulfides and halides of metals. Statement II : It consists of plosts of $\Delta_f H^0$ vs for formation of oxides of Clements. In the light of the above statements, choose the most appropriate answer from the options given below : [13-Apr-2023 shift 2]

Options:

A. Statement I is correct but Statement II is incorrect

B. Statements I is incorrect but Statement II is correct

C. Both Statement I and Statement II are incorrect

D. Both Statement I and Statement II are correct

Answer: A

Solution:

Solution:
Statement I is correct, Ellingham diagram can be constructed for formation of oxides, sulphides and halides of metals. (Ref: NCERT)
Statement II is incorrect because Ellingham diagram consists of $\Delta_f G^0$ vs T for formation of oxides of elements.

Question22

Which one of the following is not an example of calcination? [15-Apr-2023 shift 1]

Options:

A. $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

B. $2\,PbS + 3O_2 \xrightarrow{\Delta} 2\,PbO + 2SO_2$

C. $CaCO_3 \cdot MgCO_3 \xrightarrow{\Delta} CaO + MgO + 2CO_2$

D. $Fe_2O_3 \cdot xH_2O \xrightarrow{\Delta} Fe_2O_3 + xH_2O$

Answer: B

Solution:

Solution:
Calcination is the process in which the ore of the metal is heated to high temperature in the absence or limited supple of air or oxygen. Anhydrous iron is formed during iron calcination.
$Fe_2O_3 \cdot xH_2O \xrightarrow{\Delta} Fe_2O_3 + xH_2O$

Question23

List-I		List -II	
(A)	Sphalerite	(I)	$FeCO_3$
(B)	Calamine	(II)	PbS
(C)	Galena	(III)	$ZnCO_3$
(D)	Siderite	(IV)	ZnS

Choose the most appropriate answer from the options given below : [24-Jun-2022-Shift-1]

Options:

A. (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

B. (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

C. (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

D. (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Answer: A

Solution:

Solution:

Question24

Which of the following chemical reactions represents Hall-Heroult Process?
[24-Jun-2022-Shift-2]

Options:

A. $Cr_2O_3 + 2\,Al \rightarrow Al_2O_3 + 2\,Cr$

B. $2Al_2O_3 + 3C \rightarrow 4\,Al + 3CO_2$

C. $FeO + CO \rightarrow Fe + CO_2$

D. $2[Au(CN)_2]^-_{(aq)} + Zn(s) \rightarrow 2\,Au(s) + [Zn(CN_4)]^{2-}$

Answer: B

Solution:

Solution:
Hall-Herault process is used for the extraction of aluminium by electrolysis molten Al_2O_3

$2Al_2O_3 + 3C \rightarrow 4\,Al + 3CO_2$

Question25

(a) Baryte, (b) Galena, (c) Zinc blende and (d) Copper pyrites. How many of these minerals are sulphide based?
[24-Jun-2022-Shift-2]

Answer: 3

Solution:

Solution:
Baryte : $BaSO_4$
Galena :PbS
Zinc blende: ZnS
Copper pyrites : $CuFeS_2$
Of the given minerals, only 3 are sulphide based.

Question26

Leaching of gold with dilute aqueous solution of NaCN in presence of oxygen gives complex [A], which on reaction with zinc forms the elemental gold and another complex [B]. [A] and [B], respectively are :
[25-Jun-2022-Shift-1]

Options:

A. $[Au(CN)_4]^-$ and $[Zn(CN)_2(OH)_2]^{2-}$

B. $[Au(CN)_2]^-$ and $[Zn(OH)_4]^{2-}$

C. $[Au(CN)_2]^-$ and $[Zn(CN)_4]^{2-}$

D. $[Au(CN)_4]^{2-}$ and $[Zn(CN)_6]^{4-}$

Answer: C

Solution:

Solution:
In the metallurgy of gold

$$4\,Au + 8CN^- + 2H_2O + O_2 \rightarrow 4[Au(CN)_2]^- + 4OH^-$$

$$2[Au(CN)_2]^- + Zn \rightarrow [Zn(CN)_4]^{2-} + 2\,Au$$

Question27

Given below are two statements.
Statement I : During electrolytic refining, blister copper deposits precious metals.
Statement II : In the process of obtaining pure copper by electrolysis method, copper blister is used to make the anode.
In the light of the above statements, choose the correct from the options given below.
[25-Jun-2022-Shift-2]

Options:

A. Both Statement I and Statement II are true.

B. Both Statement I and Statement II are false.

C. Statement I is true but Statement II is false.

D. Statement I is false but Statement II is true.

Answer: A

Solution:

Solution:
Copper is refined using an electrolytic method. Anodes are of impure copper and pure copper strips are taken as cathode.
Impurities from the blister copper deposit as anode mud which contains antimony, selenium, tellurium, silver, gold and platinum.
Hence both statements are true

Question28

Given below are two statements :
Statement I : According to Ellingham diagram, any metal oxide with higher ΔG° is more stable than the one with lower ΔG°.
Statement II : The metal involved in the formation of oxide placed lower in the Ellingham diagram can reduce the oxide of a metal placed higher in the diagram.
In the light of the above statements, choose the most appropriate answer from the options given below :
[26-Jun-2022-Shift-1]

Options:

A. Both Statement I and Statement II are correct.

B. Both Statement I and Statement II are incorrect.

C. Statement I is correct but Statement II is incorrect.

D. Statement I is incorrect but Statement II is correct.

Answer: D

Solution:

Solution:
Ellingham diagram is plot of $\Delta GvvT$. The criterion for the feasibility of a thermal reduction is that at a given temperature Gibbs energy change of a reaction must be negative. The change in Gibbs energy, ΔG for any process at any specified temperature is given by the equation
$\Delta G = \Delta H - T \Delta S$
where $\Delta H =$ enthalpy change and $\Delta S =$ entropy change
According to the ellingham diagram, any metal oxide with higher ΔG° has a tendency of getting reduced by the metal whose metal oxide has lower value of ΔG°.

Question29

The role of depressants in 'Froth Flotation method' is to [26-Jun-2022-Shift-2]

Options:

A. selectively prevent one component of the ore from coming to the froth.

B. reduce the consumption of oil for froth formation.

C. stabilize the froth.

D. enhance non-wettability of the mineral particles.

Answer: A

Solution:

Solution:
Depressant prevent one component from coming to the froth.
For eg., in Galena ore, the depressant (NaCN) prevents impurity (ZnS) from coming to the froth.

Question30

Choose the correct answer from the options given below :

List - I		List - II	
(A)	Concentration of gold ore	(I)	Aniline
(B)	Leaching of alumina	(II)	NaOH
(C)	Froth stabiliser	(III)	SO_2
(D)	Blister copper	(IV)	NaCN

[27-Jun-2022-Shift-1]

Options:

A. (A) – (IV), (B) – (III), (C) – (II), (D) – (I)

B. (A) - (IV), (B) - (II), (C) - (I), (D) - (III)

C. (A) − (III), (B) − (II), (C) − (I), (D) − (IV)

D. (A) − (II), (B) − (IV), (C) − (III), (D) − (I)

Answer: B

Solution:

Solution:
Gold is concentrated by cyanidation
Leaching of alumina is done by $NaOH$
Froth stabiliser is aniline
Blister copper has condensed SO_2 on the surface

Question31

Statement I : Leaching of gold with cyanide ion in absence of air / O_2 leads to cyano complex of Au(III).
Statement II : Zinc is oxidized during the displacement reaction carried out for gold extraction.
In the light of the above statements, choose the correct answer from the options given below.
[27-Jun-2022-Shift-2]

Options:

A. Both Statement I and Statement II are correct.

B. Both Statement I and Statement II are incorrect.

C. Statement I is correct but Statement II is incorrect.

D. Statement I is incorrect but Statement II is correct.

Answer: D

Solution:

Solution:
Leaching of gold with cyanide ion is done in presence of air/ O_2 leading to cyano complex $[Au(CN)_2]^-$where Au is in $+1$ oxidation state.

$$4\,Au(s) + 8CN^-(aq) + 2H_2O(aq) + O_2(g) \longrightarrow 4[Au(CN)_2]_{aq}^- + 4OH^-$$

$2[Au(CN)_2]^-_{(aq)} + Zn(s) \longrightarrow [Zn(CN)_4]^{2-}_{(aq)} + 2\,Au(s)$

Zinc is oxidised from (0) to +2 oxidation state during displacement reaction carried out for gold extraction.

Question32

Given are two statements one is labelled as Assertion A and other is labelled as Reason R.
Assertion A: Magnesium can reduce Al_2O_3 at a temperature below 1350°C, while above 1350°C, while above 1350°C aluminium can reduce MgO.
Reason R : The melting and boiling points of magnesium are lower than those of aluminium.
n light of the above statements, choose most appropriate answer from the options given below:
[28-Jun-2022-Shift-1]

Options:

A. Both A and R are correct, and R is correct explanation of A.

B. Both A and R are correct, but R is NOT the correct explanation of A.

C. A is correct R is not correct.

D. A is not correct, R is correct.

Answer: B

Solution:

Solution:

Magnesium can reduce Al_2O_3 at a temperature below $1350^\circ C$ while above $1350^\circ C$ aluminium can reduce MgO because below $1350^\circ C$ ΔG of MgO (formation) is more negative and above $1350^\circ C \Delta G$ of Al_2O_3 (formation) is more negative.

The melting and boiling point of magnesium are lower than those of aluminium but it is not the correct reason.

Question33

In the metallurgical extraction of copper, following reaction is used:
$FeO + SiO_2 \rightarrow FeSiO_3$
FeO and $FeSiO_3$ respectively are.
[28-Jun-2022-Shift-2]

Options:

A. gangue and flux.

B. flux and slag.

C. slag and flux.

D. gangue and slag.

Answer: D

Solution:

Solution:
$FeO + SiO_2 \rightarrow FeSiO_3$
FeO = Gangue and $FeSiO_3$ = Slag

Question34

In isolation of which one of the following metals from their ores, the use of cyanide salt is not commonly involved?
[29-Jun-2022-Shift-1]

Options:

A. Zinc

B. Gold

C. Silver

D. Copper

Answer: D

Solution:

Solution:
In the extraction of Silver and Gold, $NaCN$ is used to leach the metal.

$$4\,Au(s) + 8CN^{-}(aq) + 2H_2O + O_2(g) \rightarrow 4[Au(CN)_2]^{-}(aq) + 4OH^{-}(aq)$$

$$2[Au(CN)_2]^{-}(aq) + Zn \rightarrow 2\,Au(s) + [(Zn(CN)_4)]^{2-}(aq)$$

In case of ore containing ZnS and PbS, the depressant used is $NaCN$.

Question35

Match List I with List II.

List-I		List-II	
(A)	Siderite	(I)	$FeCO_3$
(B)	Malachite	(II)	$CuCO_3 \cdot Cu(OH)_2$
(C)	Sphalerite	(III)	ZnS
(D)	Calamine	(IV)	$ZnCO_3$

Choose the correct answer from the options given below : [29-Jun-2022-Shift-2]

Options:

A. A-I, B-II, C-III, D-IV

B. A-III, B-IV, C-II, D-I

C. A-IV, B-III, C-I, D-II

D. A-I, B-II, C-IV, D-III

Answer: A

Solution:

Solution:
Siderite $\rightarrow FeCO_3$
Malachite $\rightarrow CuCO_3 \cdot Cu(OH)_2$
Sphalerite $\rightarrow ZnS$
Calamine $\rightarrow ZnCO_3$

Question36

The compound(s) that is (are) removed as slag during the extraction of copper is:
(A) CaO
(B) FeO
(C) Al_2O_3
(D) ZnO
(E) NiO
Choose the correct answer from the options given below:
[25-Jul-2022-Shift-1]

Options:

A. (C), (D) only

B. (A), (B), (E) only

C. (A), (B) only

D. (B) only

Answer: D

Solution:

Solution:
The compound(s) that are removed as a slag during the extraction of copper is :

$$FeS \xrightarrow{O_2/SiO_2} FeSiO_3 + SO_2$$

∴ Only iron oxide (FeO) formed slag during extraction of copper.

Question37

Given below are two statements.
Statement I : Pig iron is obtained by heating cast iron with scrap iron.
Statement II : Pig iron has a relatively lower carbon content than that of cast iron.
In the light of the above statements, choose the correct answer from

the options given below.
[25-Jul-2022-Shift-2]

Options:

A. Both Statement I and Statement II are correct.

B. Both Statement I and Statement II are not correct.

C. Statement I is correct but Statement II is not correct.

D. Statement I is not correct but Statement II is correct.

Answer: B

Solution:

Solution:
Cast iron is made by melting pig iron with scrap iron and coke using hot air blast.
Hence Statement-I is incorrect
But Pig iron has relatively more carbon content
Hence statement-II is incorrect

Question38

Refining using liquation method is the most suitable for metals with:
[26-Jul-2022-Shift-1]

Options:

A. Low melting point

B. High boiling point

C. High electrical conductivity

D. Less tendency to be soluble in melts than impurities

Answer: A

Solution:

Solution:
Refining using liquation method is the most suitable for metals with low melting point

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Question39

The metal that is not extracted from its sulfide ore is [26-Jul-2022-Shift-2]

Options:

A. Aluminium

B. Iron

C. Lead

D. Zinc

Answer: A

Solution:

Solution:
Aluminium is not extracted from sulphide ore. It is usually extracted from bauxite ore, leaching of bauxite ore is done followed by electrolytic reduction.
Al is extracted from $Al_2O_3 \cdot 2H_2O$ i.e., Bauxite ore

Question40

Which of the following methods are not used to refine any metal?
A. Liquation
B. Calcination
C. Electrolysis
D. Leaching
E. Distillation
Choose the correct answer from the options given below: [27-Jul-2022-Shift-1]

Options:

A. B and D only

B. A, B, D and E only

C. B, D and E only

D. A, C and E only

Answer: A

Solution:

Solution:
Leaching and calcination are the processes which are involved in the extraction of the metals.
Liquation, Electrolytic refining, Distillation are used in the refining or purification of metal.

Question41

Among the following ores Bauxite, Siderite, Cuprite, Calamine, Haematite, Kaolinite, Malachite, Magnetite, Sphalerite, Limonite, Cryolite, the number of principal ores if iron is _____. [27-Jul-2022-Shift-2]

Answer: 4

Solution:

Solution:
Bauxite $-AlO_x(OH)_{3-2x}$(where $0 < x < 1$)
✓ Siderite $-FeCO_3$
Cuprite $-Cu_2O$
Calamine - $ZnCO_3$
✓Fe_2O_3
Haematite- $-Al_2(OH_4)_4Si_2O_5$
Kaolinite $-CuCO_3 \cdot Cu_2(OH)_2$
Malachite $-Fe_3O_4$
✓ Magnetite $-ZnS_2$
Sphalerite $-Fe_2O_3 \cdot 3H_2O$
✓ Limonite $-Na_3AlF_6$

Question42

Which of the reaction is suitable for concentrating ore by leaching process?
[28-Jul-2022-Shift-1]

Options:

A. $2Ch_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$

B. $Fe_3O_4 + CO \rightarrow 3\,FeO + CO_2$

C. $Al_2O_3 + 2\,NaOH + 3H_2O \rightarrow 2\,Na[Al(OH)_4]$

D. $Al_2O_3 + 6\,Mg \rightarrow 6\,MgO + 4\,Al$

Answer: C

Solution:

Solution:
Leaching involves the treatment of ore with a suitable reagent so as it make it soluble while impurities remain insoluble.
$Al_2O_3 + 2\,NaOH + 3H_2O \rightarrow 2\,Na[Al(OH)_4]$
Soluble complex

Question43

In metallurgy the term "gangue" is used for :
[29-Jul-2022-Shift-1]

Options:

A. Contamination of undesired earthy materials.

B. Contamination of metals, other than desired metal.

C. Minerals which are naturally occurring in pure form

D. Magnetic impurities in an ore.

Answer: A

Solution:

Solution:
Earthy and undesired materials present in the ore other then the desired metal is known as gangue.

Question44

In liquation process used for tin (Sn), the metal [29-Jul-2022-Shift-2]

Options:

A. is reacted with acid.

B. is dissolved in water.

C. is brought to molten form which is made to flow on a slope.

D. is fused with NaOH

Answer: C

Solution:

Solution:
In this method a low melting metal like tin can be made to flow on a sloping surface. In this way it is separated from higher melting impurities

---

Question45

Match List-I with List-II.

	List-I		List-II
A.	Siderite	(i)	Cu
B.	Calamine	(ii)	Ca
C.	Malachite	(iii)	Fe
D.	Cryolite	(iv)	Al
		(v)	Zn

Chose the correct answer from the options given below. [26 Feb 2021 Shift 2]

Options:

A. (A-iii), (B-i), (C-v), (D-ii)

B. (A-i), (B-ii), (C-v), (D-iii)

C. (A-iii), (B-v), (C-i), (D-iv)

D. (A-i), (B-ii), (C-iii), (D-iv)

Answer: C

Solution:

Solution:
(A) Siderite ($FeCO_3$) is the ore of $Fe \Rightarrow$ (iii) of List-II.
(B) Calamine $(ZnCO_3)$ is the ore of $Zn \Rightarrow$ (v) of List - II.
(C) Malachite $[CuCO_3 \cdot Cu(OH)_2]$ is the ore of $Cu \Rightarrow$ (i) of List II.
(D) Cryolite (Na_3AlF_6) is the ore of $Al \Rightarrow$ (iv) of List II. (as well as F_2)
So, option -(c) is the correct matching.
Note Siderite and calamine are carbonate ores whereas malachite is basic carbonate ore. Cryolite is the thioxide ore.

Question46

Match List-I with List-II.

List-I(Ore)	List-II(Element present)
AKernite	(i) Tin
B. Cassiterite	(ii)Boron
C. Calamine	(iii) Fluorine
D. Cryolite	(iv) Zinc

Choose the most appropriate answer from the options given below. [26 Feb 2021 Shift 1]

Options:

A. (A-i) (B-iii) (C-iv) (D-ii)

B. (A-ii) (B-i) (C-iv) (D-iii)

C. (A-ii) (B-iv) (C-i) (D-iii)

D. (A-iii) (B-i) (C-ii) (D-iv)

Answer: B

Solution:

Solution:
(A) Kernite $(Na_2B_4O_7 \cdot 4H_2O)$ is sodium tetraborate tetrahydrate.
It is also called rasovite. Kernite is also written as
$Na_2B_4O_6(OH)_2 \cdot 3H_2O$
It is an ore of boron (ii).
(B) Cassiterite (SnO_2) is the oxide. It is an ore of tin (i).
(C) Calamine $(ZnCO_3)$ is zinc carbonate. It is also called smith sonite. It is an ore of zinc (iv).
(D) Cryolite (Na_3AlF_6) is sodium hexafluoroaluminate. It is an ore of fluorine (iii).
So, correct match is (A)-(ii), (B)-(i), (C)-(iv), (D)-(iii).

Question47

Ellingham diagram is a graphical representation of [25 Feb 2021 Shift 1]

Options:

A. ΔG vs T

B. ΔH vs T

C. ΔG vs p

D. $(\Delta G - T\ \Delta S)$vsT

Answer: A

Solution:

Solution:
Ellingham diagram is the graphical representation of ΔG vs T. For the formation of oxides of elements similar diagrams can also be constructed for sulphides and halides. Such diagrams can help us in pridicting the feasibility of thermal reduction of an ore.

Question48

Match List-I with List-II.

List-I (Metal)	List-II (Ores)
A. Aluminium	1. Siderite
B. Iron	2. Calamine
C. Copper	3. Kaolinite
D. Zinc	4. Malachite

Choose the correct answer from the options given below. [24 Feb 2021 Shift 2]

Options:

A. (A-4), (B-3), (C-2), (D-1)

B. (A-2), (B-4), (C-1), (D-3)

C. (A-1), (B-2), (C-3), (D-4)

D. (A-3), (B-1), (C-4), (D-2)

Answer: D

Solution:

Solution:
(A) Siderite $\rightarrow F eCO_3$
Siderite is a mineral composed of iron (II) carbonate $(F eCO_3)$.
(B) Calamine $\rightarrow Z nCO_3$
Calamine is used to treat mild itchiness and contain zinc in its formula.
(C) Kaolinite $\rightarrow Al_2(OH)_4 \cdot Si_2O_5$
Kaolinite is a type of clay found in nature. It is a dioctahedral pyrosilicate clay containing Si_2O_5.
(D) Malachite $\rightarrow Cu(OH)_2 \cdot CuCO_3$
Malachite is a copper carbonate hydroxide mineral, with the formula $Cu_2CO_3(OH)_2$.

Question49

The method used for the purification of indium is [25 Feb 2021 Shift 2]

Options:

A. van-Arkel method

B. liquation

C. zone refining

D. vapour phase refining

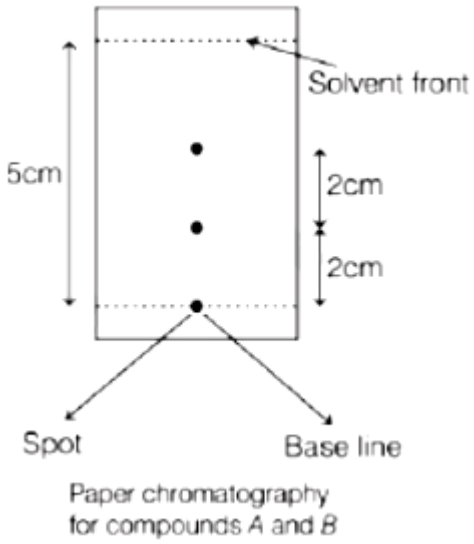
Answer: C

Solution:

Solution:
Zone refining is used for the purification of indium.

Question50

Using the provided information in the following paper chromatogram.



Paper chromatography
for compounds *A* and *B*

The calculated R_f value of A $\times 10^{-1}$ [25 Feb 2021 Shift 1]

Answer: 4

Solution:

Solution:
The retention factor (R_f) value can be used to identify the components of a mixture (solutes).

$$R_f = \frac{\text{Distance travelled by solute}}{\text{Distance travelled by solvent front}}$$

On chromatogram, distance travelled by compound is 2cm.
Distance travelled by solvent $= 5\text{cm}$

$$\therefore \ R_f = \frac{2}{5} = 4 \times 10^{-1}$$

Question51

Which of the following ore is concentrated using group 1 cyanide salt?
[24 Feb 2021 Shift 1]

Options:

A. Sphalerite

B. Calamine

C. Siderite

D. Malachite

Answer: A

Solution:

Solution:
Sphalerite ore : ZnS Calamine ore : $ZnCO_3$ Siderite ore : $FeCO_3$ Malachite ore : $Cu(OH)_2 \cdot CuCO_3$ In the froth flotation process (concentrated method), depressant helps to separates two sulphide ores by selective preventation of froth formation by one ore and allowing the other to come into froth.
For example, in case of an ore containing ZnS and PbS, $NaCN$ is used as a depressant.
$NaCN$ reacts with ZnS to form $Na_2[Zn(CN)_4]$, so it prevents ZnS from coming to the froth but allow PbS to come with the froth.

Question52

Match List-I and List-II.

List-I	List-II
A. Haematite	1. $Al_2O_3 \cdot \times H_2O$
B. Bauxite	2. Fe_2O_3
C. Magnetite	3. $CuCO_3 \cdot Cu(OH)_2$
D. Malachite	4. Fe_3O_4

Choose the correct answer from the options given below. [17 Mar 2021 Shift 2]

Options:

A. A-2 ,B-3 ,C-1 ,D-4

B. A-4 ,B-1 ,C-2 ,D-3

C. A-1 ,B-3 ,C-2 ,D-4

D. A-2 ,B-1 ,C-4 ,D-3

Answer: D

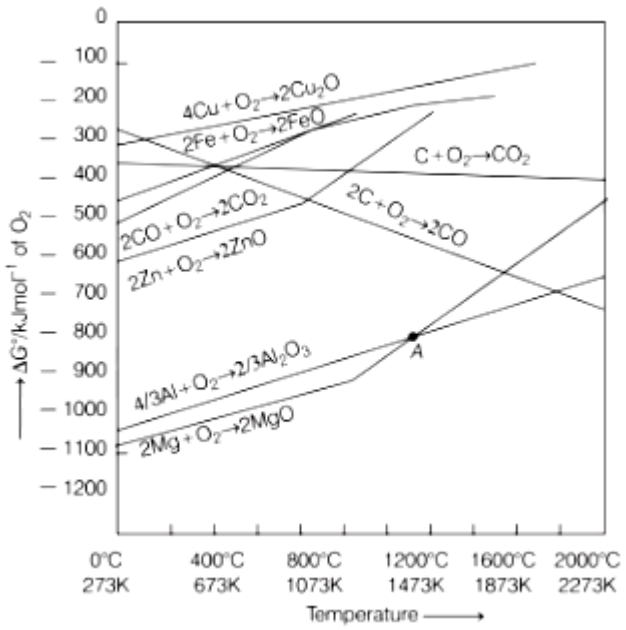
Solution:

Solution:
Correct matches are A-2, B-1, C-4, D-3.

Ore	Formula
Haematite	Fe_2O_3
B. Bauxite	$Al_2O_3 \cdot \times H_2O$
C. Magnetite	Fe_3O_4
D. Malachite	$CuCO_3 \cdot Cu(OH)_2$

Question53

The point of intersection and sudden increase in the slope, in the diagram given below, respectively, indicates.



[17 Mar 2021 Shift 1]

Options:

A. $\Delta G = 0$ and melting or boiling point of the metal oxide

B. $\Delta G > 0$ and decomposition of the metal oxide

C. $\Delta G < 0$ and decomposition of the metal oxide

D. $\Delta G = 0$ and reduction of the metal oxide

Answer: A

Solution:

Solution:
At intersection point $\Delta G = 0$, which represents the equilibrium
point and sudden increase in slope is due to melting or boiling point of the metal oxide.
According to Gibb's free rule,
$\Delta G = \Delta H - T\ \Delta S$
where, ΔH = change in enthalpy ΔG = change in Gibb's free energy
ΔS = change in entropy
We can see that ΔG is dependent on ΔS.
So, when the phase changes ($s \rightarrow l$ or $l \rightarrow g$) the entropy increases and hence, slope increases.

Question54

Match List-I with List-II.

List-I	List-II
A. Mercury	1. Vapour phase refining
B. Copper	2. Distillation refining
C. Silicon	3. Electrolytic refining
D. Nickel	4. Zone refining

Choose the most appropriate answer the option given below [18 Mar 2021 Shift 2]

Options:

A. A- 1 ,B- 4,C- 2,D- 3

B. A- 2 ,B- 3,C- 1,D- 4

C. A- 2 ,B- 3,C- 4,D- 1

D. A- 24 ,B- 4,C- 3,D- 1

Answer: C

Solution:

Solution:
(A) The pyrometallurgical extraction of mercury from its ore is essentially a distillation process.
(B) The purification of metals by electrolysis is commonly applied to copper. A large piece of impure copper is used as the anode with a thin strip of pure copper as the cathode.
(C) Zone refining is a very useful method to get metals with very high purity such as silicon and germanium. It is also referred as zone melting.
(D) Mond's process for extraction of nickel and van-Arkel method for preparing ultra-pure titanium are based on the principle of vapour phase refining.
Hence, correct match is (A-2), (B-3), (C-4), (D-1).

Question55

The chemical that is added to reduce the melting point of the reaction mixture during the extraction of aluminium is [18 Mar 2021 Shift 1]

Options:

A. cryolite

B. bauxite

C. calamine

D. kaolite

Answer: A

Solution:

Solution:
In the extraction of aluminium by electrolytic process, cryolite is added to reduce the melting point of the mixture. The use of cryolite reduces some of the energy cost involved in extracting aluminium.
Reactions involved in the electrolysis are given below

$Al_2O_3 \rightleftharpoons 2Al^{+3} + 3O^{-2}$

$Na_3AlF_6 \rightleftharpoons 3Na^{+} + Al^{+3} + 6F$

$CaF_2 \rightleftharpoons Ca^{+2} + 2F$

At cathode

$Al^{3+} + 3e^{-} \longrightarrow Al$ (reduction)

At anode

$C + O^{2-} \longrightarrow CO + 2e^{-}$(oxidation)

$C + 2O_2^{-} \longrightarrow CO_2 + 4e^{-}$

Question56

Which of the following reduction reaction cannot be carried out with coke?
[16 Mar 2021 Shift 2]

Options:

A. $Al_2O_3 \longrightarrow Al$

B. $ZnO \longrightarrow Zn$

C. $Fe_2O_3 \longrightarrow Fe$

D. $Cu_2O \longrightarrow Cu$

Answer: A

Solution:

Solution:
Reduction of alumina (Al_2O_3) to Al cannot be achieved by carbon reduction. As aluminium has greater reactivity with oxygen than carbon and hence cannot be reduced by carbon. So, electrometallurgy (Hall-Heroult process) is used to get Al.

$$2Al_2O_3(I) + 3C \longrightarrow 4Al + 3CO_2$$

Rest oxides can be reduced by coke as follows

$$Fe_2O_3 + 3C \longrightarrow 2Fe + 3CO$$

$$Cu_2O + C \longrightarrow 2Cu + CO$$

$$ZnO + C \xrightarrow{\text{Coke, } 1673K} Zn + CO$$

Question57

The statement that is INCORRECT about Ellingham diagram is [27 Jul 2021 Shift 1]

Options:

A. provides idea about the reaction rate.

B. provides idea about free energy change.

C. provides idea about changes in the phases during the reaction.

D. provides idea about reduction of metal oxide.

Answer: A

Solution:

Solution:
Ellingham diagram is a plot between ΔG° and T and does not give any information regarding rate of reaction

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Question58

Sulphide ion is soft base and its ores are common for metals.
(a) Pb
(b) Al
(c) Ag

(d) Mg

Choose the correct answer from the options given below :
[22 Jul 2021 Shift 2]

Options:

A. (a) and (c) only

B. (a) and (d) only

C. (a) and (b) only

D. (c) and (d) only

Answer: A

Solution:

Solution:
Pb and Ag commonly exist in the form of sulphideore like PbS (galena) and Ag_2S (Argentite)
'Al' is mainly found in the form of oxide orewhereas 'Mg' is found in the form of halide ore.

Question59

Given below are two statements. One is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : Sharp glass edge becomes smooth on heating it upto its melting point.
Reason R : The viscosity of glass decreases on melting.
Choose the most appropriate answer from the options given below.
[20 Jul 2021 Shift 1]

Options:

A. A is true but R is false

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is false but R is true.

D. Both A and R are true and R is the correct explanation of A.

Answer: B

Solution:

Solution:
On heating the glass, it melts and takes up rounded shape at the edges, which has minimum surface area. This is due to the property of surface tension of liquids and not due to decrease in viscosity.
Viscosity generally decreases as the temperature increases.
Hence, both A and R are true but R is not the correct explanation of A.

Question60

The addition of silica during the extraction of copper from its sulphide ore :-
[27 Jul 2021 Shift 2]

Options:

A. converts copper sulphide into copper silicate

B. converts iron oxide into iron silicate

C. reduces copper sulphide into metallic copper

D. reduces the melting point of the reactionmixture

Answer: B

Solution:

Solution:
Silica is used to remove FeO impurity from the ore of copper

$$FeO + SiO_2 \rightarrow \underset{\text{iron silicate(Slag)}}{FeSiO_3}$$

Question61

Match List I with List II : (Both having metallurgical terms)

	List-I		List-II
(a)	Concentration of Ag ore	(i)	Reverberatory furnace
(b)	Blast furnace	(ii)	Pig iron
(c)	Blister copper	(iii)	Leaching with dilute NaCN solution
(d)	Froth floatation method	(iv)	Sulfide ores

Choose the correct answer from the options given below : [25 Jul 2021 Shift 2]

Options:

A. (a)–(iii), (b)–(ii), (c)–(i), (d)–(iv)

B. (a)–(iii), (b)–(iv), (c)–(i), (d)–(ii)

C. (a)–(iv), (b)–(i), (c)–(iii), (d)–(ii)

D. (a)–(iv), (b)–(iii), (c)–(ii), (d)–(i)

Answer: A

Solution:

Solution:
(a) Concentration of Ag is performed by leaching with dilute NaCN solution
(b) Pig iron is formed in blast furnace
(c) Blister Cu is produced in Bessemer converter
(d) Froth floatation method is used for sulphide ores.
Note : During extraction of Cu reverberatory furnace is involved

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Question62

In the leaching of alumina from bauxite, the ore expected to leach out in the process by reacting with N aOH is : [25 Jul 2021 Shift 1]

Options:

A. TiO_2

B. Fe_2O_3

C. ZnO

D. SiO_2

Answer: D

Solution:

Solution:
In bauxite impurities of Fe_2O_3, TiO_2 and SiO_2 are present, Fe_2O_3 and TiO_2 are basic oxides therefore does not reacts with or dissolve in NaOH whereasSiO_2 is acidic oxide it gets dissolve in NaOH , hence leach out
$SiO_2 + 2NaOH \rightarrow Na_2SiO_3(aq.) + H_2O$

Question63

Consider two chemical reactions (A) and (B) that take place during metallurgical process :

(A) $\mathbf{ZnCO_{3(s)} \xrightarrow{\Delta} ZnO_{(s)} + CO_{2(g)}}$

(B) $\mathbf{2ZnS_{(s)} + 3O_{2(g)} \xrightarrow{\Delta} 2ZnO_{(s)} + 2SO_{2(g)}}$

The correct option of names given to them respectively is : [20 Jul 2021 Shift 2]

Options:

A. (A) is calcination and (B) is roasting

B. Both (A) and (B) are producing same product so both are roasting

C. Both (A) and (B) are producing same product so both are calcination

D. (A) is roasting and (B) is calcination

Answer: A

Solution:

Solution:

(A) $ZnCO_3(S) \xrightarrow{\Delta} ZnO(s) + CO_2(g)$

Heating in absence of oxygen in calcination.

(B) $2ZnS(s) + 3O_2(g) \rightarrow 2ZnO(g) + 2SO_2(g)$

heating in presence of oxygen in roasting

Hence (A) is calcination while (B) in roasting.

Question64

Match List-I with List-II.

List-I (Name ofore/mineral)	List-II (Chemical formula)
A. Calamine	1. ZnS
B. Malachite	2. $FeCO_3$
C.Siderite	3. $ZnCO_3$
D. Sphalerite	4. $CuCO_3 . Cu(OH)_2$

Choose the most appropriate answer from the options given below [27 Aug 2021 Shift 2]

Options:

A. A-3 B-4 C-2 D-1

B. A-3 B-4 C-1 D-2

C. A-4 B-3 C-1 D-2

D. A-3 B-2 C-4 D-1

Answer: A

Solution:

Solution:

Ore and their chemical formula are given below

(A) Calamine (3) $ZnCO_3$

(B) Malachite (4) $CuCO_3 . Cu(OH)_2$

(C) Siderite (2) $FeCO_3$

(D) Sphalerite (1)ZnS
Therefore, the correct match is A → 3, B → 4, C → 2, D → 1.

Question65

Given below are two statements.
Statement I The choice of reducing agent for metals extraction can be made by using Ellingham diagram, a plot of ΔG vs temperature.
Statement II The value of ΔS increases from left to right in Ellingham diagram. In the light of the above statements, choose the most appropriate answer from the options given below.
[26 Aug 2021 Shift 1]

Options:

A. Both statement I and statement II are true.

B. Statement I is false but statement II is true.

C. Both statement I and statement II are false.

D. Statement I is true but statement II is false.

Answer: D

Solution:

Solution:
The Ellingham diagram, a plot of ΔG vs temperature is used to choose reducing agents for metal extraction. Any element will reduce the oxide of other metal if it lie above that element in Ellingham diagram because the free energy change will becomemore negative.
Hence, statement I is true.
In Ellingham diagram the value of entropy change (ΔS) decreases from left to right.
Hence, statement II is false.

Question66

Given below are two statements.
One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) Aluminium is extracted from bauxite by the

electrolysis of molten mixture of Al_2O_3 with cryolite.

Reason (R) The oxidation state of Al in cryolite is +3.
In the light of the above statements, choose the most appropriate answer from the options given below.
[31 Aug 2021 Shift 1]

Options:

A. (A) is true but (R) is false

B. (A) is false but (R) is true.

C. Both (A) and (R) are true and (R) is the correct explanation of (A).

D. Both (A) and (R) are true but (R) is not the correct explanation of (A).

Answer: D

Solution:

Solution:
Aluminium is reactive metal. So, aluminium is extracted by electrolysis of alumina with molten mixture of cryolite. Hence, Assertion is true.
Cryolite has molecular formula Na_3AlF_6.
Oxidation state of Al in Na_3AlF_6
$3 \times (+1) + Al + 6 \times (-1) = 0$
$3 + Al - 6 = 0$
$\Rightarrow$ Al = + 3
Hence, Reason is true.
Oxidation state cannot explain that Al is extracted from bauxite by electrolysis.
$\therefore$ The correct option is (d).

Question67

Given below are two statements.
Statement I Sphalerite is a sulphide ore of zinc and copper glance is a sulphide ore of copper.
Statement II It is possible to separate two sulphide ores by adjusting proportion of oil to water or by using 'depressants' in a froth flotation method.
Choose the most appropriate answer from the options given below.
[26 Aug 2021 Shift 2]

Options:

A. Statement I is true but statement II is false.

B. Both statement I and statement II are true.

C. Statement I is false but statement II is true.

D. Both statement I and statement II are false.

Answer: B

Solution:

Solution:
Both statements I and II are true.
Sphalerite –ZnS (Sulphide ore of zinc).
Copper glance – Cu_2S (Sulphide ore of copper).
It is possible to separate two sulphide ores by adjusting the proportion of oil to water or by using depressants.

Question68

In the electrolytic refining of blister copper, the total number of main impurities from the following, removed as anode mud is ..... Pb, Sb, Se, Te, Ru, Ag, Au and Pt
[31 Aug 2021 Shift 2]

Answer: 6

Solution:

Solution:
Impurities from blister copper deposited as anode mud contain antimony, selenium, tellurium, silver, gold and platinum.
These are basically less active metals which cannot be oxidised at anode.

Question69

Which refining process is generally used in the purification of low melting metals ?
[27 Aug 2021 Shift 1]

Options:

A. Chromatographic method

B. Liquation

C. Electrolysis

D. Zone refining

Answer: B

Solution:

Solution:
Liquation is used for separating metals which have lower melting points than the impurities. The process is done by keeping the metal to be refined in a sloping surface of a furnace. The temperature will be maintained above the melting point of metals. Thus, pure metal melts and moves down.

Question70

Calamine and malachite, respectively, are the ores of
[1 Sep 2021 Shift 2]

Options:

A. nickel and aluminium

B. zinc and copper

C. copper and iron

D. aluminium and zinc

Answer: B

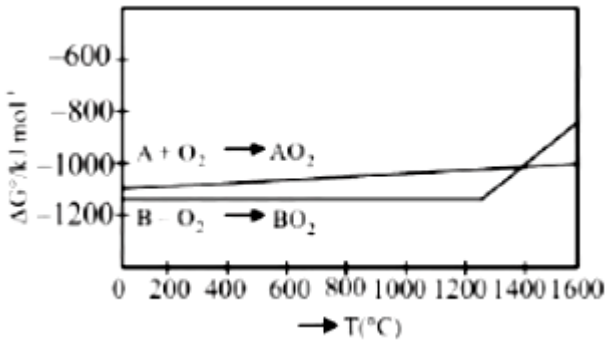
Solution:

Solution:

Calamine is the ore of Zn and is found as zinc carbonate ($ZnCO_3$), where malachite is the ore of copper and is found as copper hydroxide [Cu $(OH)_2$] and copper carbonate [CuCO3].

Question71

According to the following diagram, A reduces BO_2 when the temperature is:



[Jan. 09,2020(I)]

Options:

A. $<1400^{\circ}C$

B. $>1400^{\circ}C$

C. $>1200^{\circ}C$but $< 1400^{\circ}C$

D. $<1200^{\circ}C$

Answer: B

Solution:

Solution:

$A + BO_2 \longrightarrow B + AO_2$

$\Delta G = -ve$

when the temperature is above $1400^{\circ}C$, A reduces BO_2.

Question72

Among the reactions (a) −(d), the reaction(s) that does/ do not occur in the blast furnace during the extraction of iron is/are:

(a) $CaO + SiO_2 \rightarrow CaSiO_3$

(b) $3Fe_2O_3 + CO \rightarrow 2Fe_3O_4 + CO_2$

(c) $FeO + SiO_2 \rightarrow FeSiO_3$

(d) $FeO \rightarrow Fe + \frac{1}{2}O_2$

[Jan .08,2020(II)]

Options:

A. (a)

B. (a) and (d)

C. (c) and (d)

D. (d)

Answer: C

Solution:

Solution:
During the reduction of Fe_2O_3 to FeO
Fe_2O_3 is first reduced to Fe_3O_4 and then to FeO,
thus follows reaction (b).
In order to remove impurity of the ore limestone is added along with ore which is decomposed to CaO.
The resulted CaO reacts with SiO_2 to form $CaSiO_3$ i.e. follows reaction (a) and is removed as a slag.

Question73

The refining method used when the metal and the impurities have low and high melting temperatures, respectively, is:
[Jan .07,2020(II)]

Options:

A. liquation

B. vapour phase refining

C. zone refining

D. distillation

Answer: A

Solution:

Solution:
Liquation method is used for purification of metals having low melting point.

Question74

The purest form of commercial iron is:
[Jan 07,2020 (I)]

Options:

A. pig iron

B. wrought iron

C. cast iron

D. scrap iron and pig iron

Answer: B

Solution:

Solution:
Wrought iron is purest form of commercial iron.

Question75

The INCORRECT statement is:
[Sep. 06,2020(I)]

Options:

A. bronze is an alloy of copper and tin.

B. cast iron is used to manufacture wrought iron.

C. german silver is an alloy of zinc, copper and nickel.

D. brass is an alloy of copper and nickel.

Answer: D

Solution:

Solution:
(a) Bronze contains $88-96$ % Cu and $4-12$ % Sn.
(b) Cast iron is used to make wraught iron.
(c) German silver contains 50 % Cu, 30 % Zn & 20 % Ni.
(d) Brass contains 70 % Cu and 30 % Zn.

Question76

For a reaction, $4M(s) + nO_2(g) \rightarrow 2M_2O_n(s)$, the free en- ergy change is plotted as a function of temperature. The temperature below which the oxide is stable could be inferred from the plot as the point at which:
[Sep. 06,2020(II)]

Options:

A. the slope change from negative to positive

B. the free energy change shows a change from negative to positive value

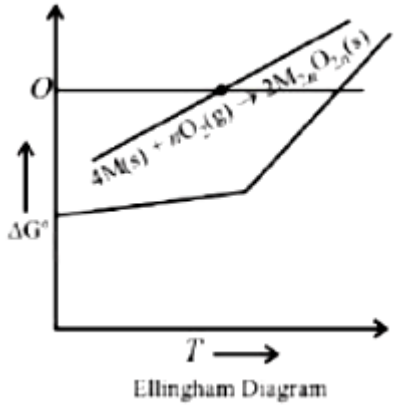
C. the slop changes from positive to negative

D. the slop changes from positive to zero

Answer: B

Solution:

Solution:



Ellingham Diagram

From graph it is evident that the temperature below which the oxide is stable, is the point at which free energy change shows a change from negative to positive.

Question77

An Ellingham diagram provides information about: |Sep. 05,2020 (I)]

Options:

A. the conditions of pH and potential under which a species is thermodynamically stable.

B. the temperature dependence of the standard Gibbs energies of formation of some metal oxides.

C. the pressure dependence of the standard electrode potentials of reduction reactions involved in the extraction of metals.

D. the kinetics of the reduction process.

Answer: B

Solution:

Solution:
Ellingham diagram is the graph of ΔG^0 vs T of any metal/element oxide. Since
$\Delta G^o = \Delta H^o - T\,\Delta\, S^\circ$
For most metal oxide formation.
$\text{metal (s)} + \text{oxygen (g)} \rightarrow \text{metal oxide (s)}$
$\Delta H^\circ = -\text{ve}; \Delta S^0 = -\text{ve}$
So, graph will be a straight line with -ve intercept and +ve slope.

Question78

Among statements (1) - (4), the correct ones are :
(1) Lime stone is decomposed to CaO during the extraction of iron from its oxides.
(2) In the extraction of silver, silver is extracted as an anionic complex.
(3) Nickel is purified by Mond's process.
(4) Zr and Ti are purified by Van Arkel method.
|Sep. 04,2020 (I)

Options:

A. (1), (2), (3) and (4)

B. (1), (3) and (4) only

C. (2), (3) and (4) only

D. (3) and (4) only

Answer: A

Solution:

Solution:
All statements are correct.
(1) $\underset{\text{Lime stone}}{CaCO_3} \xrightarrow{500-800K} CaO + CO_2$
(2) In extraction of Ag, it is extracted as an anionic complex $[Ag(CN)_2]$
(3) Ni is purified by Mond's process
$Ni + 4CO \xrightarrow{300K} Ni(CO)_4$
$Ni(CO)_4 \xrightarrow{450K} Ni + 4CO$
(4) Van Arkel method is used to purify Zr and Ti.

Question79

The element that can be refined by distillation is:
[Sep. 06,2020 (II)]

Options:

A. nickel

B. zinc

C. tin

D. gallium

Answer: B

Solution:

Solution:
Zn, Cd and Hg are purified by fractional distillation process.

Question80

Boron and silicon of very high purity can be obtained through: [Sep. 05,2020 (II)]

Options:

A. liquation

B. zone refining

C. vapour phase refining

D. electrolytic refining

Answer: B

Solution:

Solution:
Germanium (Ge), silicon (Si), boron (B), gallium (Ga) and indium (In) can be obtained in pure state by zone refining process.

Question81

Cast iron is used for the manufacture of : [Sep. 02,2020 (II)]

Options:

A. wrought iron and pig iron

B. pig iron, scrap iron and steel

C. wrought iron, pig iron and steel

D. wrought iron and steel

Answer: D

Solution:

Solution:
Cast iron is made from pig iron which is used for production of wroght iron and steel.

Question82

Match the ores (column A) with the metals (column B):

	(Column A)		(Column B)
	Ores		**Metals**
(I)	Siderite	(a)	Zinc
(II)	Kaolinite	(b)	Copper
(III)	Malachite	(c)	Iron
(IV)	Calamine	(d)	Aluminium

[Jan 11,2019 (I)

Options:

A. (I) – (a); (II) – (b); (III) – (c); (IV) – (d)

B. (I) – (c); (II) – (d); (III) – (b); (IV) – (a)

C. (I) – (c); (II) – (d); (III) – (a); (IV) – (b)

D. (I) – (b); (II) – (c); (III) – (d); (IV) – (a)

Answer: B

Solution:

Solution:
Siderite $= FeCO_3$
Calamine $= ZnCO_3$
Malachite $= CuCO_3Cu(OH)_2$
Kaolinite $= Al_2Si_2O_5(OH)_4$

Question83

The ore that contains both iron and copper is:
[Jan. 9,2019 (I)]

Options:

A. copper pyrites

B. malachite

C. dolomite

D. azurite

Answer: A

Solution:

Solution:
Amongst the given ores; copper pyrite (\CuFeS $_2$), dolomite $(MgCO_3 \cdot CaCO_3)$, malachite $[CuCO_3 \cdot Cu(OH)_2]$ azurite $[2CuCO_3 \cdot Cu(OH)_2]$ only copper pyrite contains both copper and iron.

Question84

In the Hall-Heroult process, aluminium is formed at the cathode. The cathode is made out of:
[Jan .12,2019(I)]

Options:

A. Purealuminium

B. Carbon

C. Copper

D. Platinum

Answer: B

Solution:

Solution:
In Hall-Heroult process, steel vessel with lining of carbon acts as cathode and graphite is used as anode.

Question85

The pair that does NOT require calcination is: [Jan .12,2019(II)]

Options:

A. ZnO and MgO

B. ZnO and $Fe_2O_3 \cdot xH_2O$

C. $ZnCO_3$ and CaO

D. Fe_2O_3 and $CaCO_3 \cdot MgCO_3$

Answer: A

Solution:

Solution:
Calcination involves the conversion of metal carbonates or hydrated oxides into metal oxides. ZnO and MgO are oxides, therefore, does not require calcination.

Question86

The reaction that does NOT define calcination is: [Jan .11,2019(II)]

Options:

A. $Fe_2O_3 \cdot xH_2O \xrightarrow{\Delta} Fe_2O_3 + xH_2O$

B. $2Cu_2S + 3O_2 \xrightarrow{\Delta} 2Cu_2O + 2SO_2$

C. $ZnCO_3 \xrightarrow{\Delta} ZnO + CO_2$

D. $CaCO_3 \cdot MgCO_3 \xrightarrow{\Delta} CaO + MgO + 2CO_2$

Answer: B

Solution:

Solution:
Calcination involves heating of carbonates or hydrated oxides in absence of air.
$\therefore 2Cu_2S + 3O_2 \xrightarrow{\Delta} 2Cu_2O + 2SO_2;$ is roasting (heating in a regular supply of air).

Question87

Hall-Heroult's process is given by: [Jan .10,2019(I)]

Options:

A. $Cu^{2+}(aq) + H_2(g) \rightarrow Cu(s) + 2H^+(aq)$

B. $Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr$

C. $2Al_2O_3 + 3C \rightarrow 4Al + 3CO_2$

D. $ZnO + C \xrightarrow{\text{Col ke, 1673K}} Zn + CO$

Answer: C

Solution:

Solution:
Hall Heroult's process is an electrochemical process used in extraction of Al from alumina.

Question88

The electrolytes usually used in the electroplating of gold and silver, respectively, are:
[Jan .10,2019(I)]

Options:

A. $[Au(CN)_2]^-$ and $[Ag(CN)_2]^-$

B. $[Au(CN)_2]$ and $[AgCl_2]^-$

C. $[Au(OH)_4]^-$ and $[Ag(OH)_2]$

D. $[Au(NH_3)_2]^+$ and $[Ag(CN)_2]$

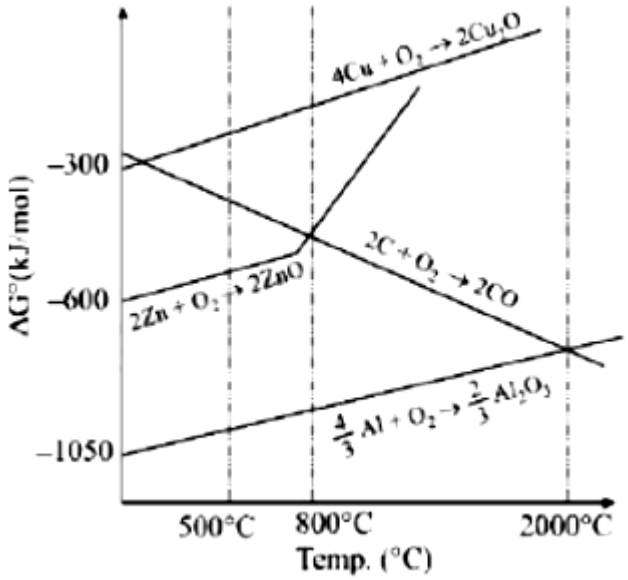
Answer: A

Solution:

Solution:
The electrolytes used in the electroplating of Au and Ag are $[Au(CN)_2]$ and $[Ag(CN)_2]^-$ respectively.

Question89

The correct statement regarding the given Ellingham diagram is:



[Jan .9,2019(II)]

Options:

A. At $1400^{\circ}C$, Al can be used for the extraction of Z n from Z nO

B. At $500^{\circ}C$, coke can be used for the extraction of Z n from 7 nO

C. Coke cannot be used for the extraction of Cu from Cu_2O

D. At $800^{\circ}C$, Cu can be used for the extraction of Z n from Z nO

Answer: A

Solution:

Solution:
In the given Ellingham diagram, the metal which has a lower value of ΔG° (more negative) can reduce a metal oxide whose curve lies above it. So, Al can reduce Z nO at $1400^{\circ}C$

Question90

The one that is not a carbonate ore is:
[April 9,2019 (II)

Options:

A. malachite

B. calamine

C. siderite

D. bauxite

Answer: D

Solution:

Solution:
Bauxite $\rightarrow Al(OH)_3$
Calamine $\rightarrow ZnCO_3$
Siderite $\rightarrow FeCO_3$
Malachite $\rightarrow CuCO_3 \cdot Cu(OH)_2$

Question91

The idea of froth floatation method came from a person X and this method is related to the process Y of ores, X and Y, respectively, are:
[April 12, 2019 (I)]

Options:

A. fisher woman and concentration

B. washer woman and concentration

C. fisher man and reduction

D. washer man and reduction

Answer: B

Solution:

Solution:
Froth floatation was discovered by washer women. It is a method of concentration of ores.

Question92

The correct statement is :
[April 12, 2019 (II)]

Options:

A. leaching of bauxite using concentrated N aOH solution gives sodium aluminate and sodium silicate.

B. the Hall-Heroult process is used for the production of aluminium and iron.

C. pig iron is obtained from cast iron.

D. the blistered appearance of copper during the metallurgical process is due to the evolution of CO_2

Answer: A

Solution:

Solution:
During metallurgy of aluminium, when bauxite (powdered ore) is treated with $NaOH$ (conc), sodium aluminate $(Na[Al(OH)_4])$ is formed and impurity (SiO_2) present in bauxite dissolves by forming sodium silicate (Na_2SiO_3)
$Al_2O_3(s) + 2NaOH(aq) + 3H_3O(I) \rightarrow 2Na[Al(OH)_4](aq)$ Cast iron is obtained from pig iron. Blister of copper is due to evolution of SO_2 Hall Heroult process is used for the production of aluminium.

Question93

The correct statement is :
[April 10, 2019 (II)]

Options:

A. aniline is a froth stabilizer.

B. zincite is a carbonate ore.

C. sodium cyanide cannot be used in the metallurgy of silver.

D. zone refining process is used for the refining of titanium.

Answer: A

Solution:

Solution:
Ti is refined by van Arkel method. Silver is leached by dilute solution of $NaCN$. Zincite is oxide ore of zinc i.e. ZnO. Aniline is a froth stabilizer.

Question94

The ore that contains the metal in the form of fluoride is:
[April 9, 2019 (I)]

Options:

A. cryolite

B. malachite

C. magnetite

D. sphalerite

Answer: A

Solution:

Solution:
Magnetite Fe_3O_4
Sphalerite ZnS
Cryolite Na_3AlF_6
Malachite $CuCO_3 \cdot Cu(OH)_2$

Question95

Assertion: For the extraction of iron, haematite ore is used.
Reason: Haematite is a carbonate ore of iron.
[April 9, 2019 (II)]

Options:

A. Only the reason is correct

B. Both the assertion and reason are correct, but the reason is not the correct explanation for the assertion.

C. Both the assertion and reason are correct and the reason is the correct explanation for the assertion.

D. Only the assertion is correct.

Answer: D

Solution:

Solution:
For the extraction of iron, haematite ore (Fe_2O_3) is used.

Question96

With respect to an ore, Ellingham diagram helps to predict the feasibility of its
[April 8, 2019 (I)]

Options:

A. Electrolysis

B. Zone refining

C. Thermal reduction

D. Vapour phase refining

Answer: C

Solution:

Solution:
Ellingham diagram gives idea that which sustance or metal is suitable for thermal reduction of oxide.

Question97

The Mond process is used for the :
[April 8, 2019 (II)]

Options:

A. purification of N i

B. extraction of Mo

C. purification of Z r and T i

D. extraction of Z n

Answer: A

Solution:

Solution:
Mond's process is used for purification of Ni

$$\underset{\text{(impure)}}{\text{N i}} + 4\text{CO} \xrightarrow{330-350\text{K}} \text{N i(CO)}_4$$

$$Ni(CO)_4 \xrightarrow{450-470K} \underset{(\text{pure})}{Ni} + 4CO\uparrow$$

Question98

Match the refining methods (Column I) with metals (Column II).

	(Column I)		(Column II)
	(Refining methods)		(Metals)
(I)	Liquation	(a)	Zr
(II)	Zone Refining	(b)	Ni
(III)	Mond Process	(c)	Sn
(IV)	Van Arkel Method	(d)	Ga

[April 10, 2019 (I)]

Options:

A. (I) – (c); (II) – (a); (III) – (b); (IV) – (d)

B. (I) – (b); (II) – (c); (III) – (d); (IV) – (a)

C. (I) – (c); (II) – (d); (III) – (b); (IV) – (a)

D. (I) – (b); (II) – (d); (III) – (a); (IV) – (c)

Answer: C

Solution:

Solution:
Liquation is used for Sn, zone refining is used for Ga, mond's process is used for refining of Ni and van Arkel method is used for Zr
So, correct match is
(I) – (c); (II) – (d); (III) – (b); (IV) – (a)

Question99

In the extraction of copper from its sulphide ore, metal is finally obtained by the oxidation of cuprous sulphide with :
[Online April 16,2018(II)]

Options:

A. SO_2

B. Fe_2O_3

C. Cu_2O

D. CO

Answer: C

Solution:

Solution:
$Cu_2S + 2Cu_2O \longrightarrow 6Cu + SO_2$ It is an example of auto-reduction, CuO_2 is reduced to copper metal.

Question100

In the leaching method, bauxite ore is digested with a concentrated solution of NaOH that produces $'X'$. When CO_2 gas is passed through the aqueous solution of ' X ', a hydrated compound 'Y' is precipitated. 'X' and 'Y' respectively are.
[Online April 15,2018(II)]

Options:

A. $Na[Al(OH)_4]$ and $Al_2(CO_3)_3 \cdot xH_2O$

B. $Al(OH)_3$ and $Al_2O_3 \cdot xH_2O$

C. $NaAlO_2$ and $Al_2(CO_3)_3 \cdot xH_2O$

D. $Na[Al(OH)_4]$ and $Al_2O_3 \cdot xH_2O$

Answer: D

Solution:

Solution:

$$\underset{\text{Alumina}}{Al_2O_3(s)} + 2NaOH(aq) + 3H_2O(l) \xrightarrow[35-36\ bar]{473-523K} \underset{\text{Sodium aluminate (X)}}{2Na[Al(OH)_4](aq)}$$

$$2Na[Al(OH)_4](aq) + CO_2(g) \longrightarrow \underset{\text{Hydrated alumina Y (Y)}}{Al_2O_3 \times H_2O(s)} + 2NaHCO_3(aq)$$

Question101

Which one of the following ores is best concentrated by froth floatation method? [2016]

Options:

A. Galena

B. Malachite

C. Magnetite

D. Siderite

Answer: A

Solution:

Solution:
Froth floatation method is mainly applicable for sulphide ores.
(1) Malachite ore : $Cu(OH)_2 \cdot CuCO_3$
(2) Magnetite ore : Fe_3O_4
(3) Siderite ore : $FeCO_3$
(4) Galena ore : PbS (Sulphide ore)

Question102

Extraction of copper by smelting uses silica as an additive to remove : [Online April 10,2016]

Options:

A. Cu_2O

B. FeS

C. FeO

D. Cu_2S

Answer: C

Solution:

Solution:
FeO is gangue and SiO_2 is flux to form slag $FeSiO_3$

$$\underset{\text{Basic impurity}}{FeO} + \underset{\text{Acidic flux}}{SiO_2} \longrightarrow \underset{\text{Slag}}{FeSiO_3}$$

Question103

Calamine is an ore of :
[Online April 11, 2015]

Options:

A. zinc

B. aluminium

C. iron

D. copper

Answer: A

Solution:

Solution:
Calamine ($ZnCO_3$) is an ore of zinc.

Question104

In the context of the Hall - Heroult process for the extraction of Al, which of the following statements is false?
[2015]

Options:

A. Al^{3+} is reduced at the cathode to form Al

B. Na_3AlF_6 serves as the electrolyte

C. CO and CO_2 are produced in this process

D. Al_2O_3 is mixed with CaF_2 which lowers the melting point of the mixture and brings conductivity

Answer: B

Solution:

Solution:
In the metallurgy of aluminium, purified Al_2O_3 is mixed with Na_3AlF_6 or CaF_2 which lowers the melting point of the mix and brings conductivity.

Question105

In the isolation of metals, calcination process usually results in :
[Online April 10,2015]

Options:

A. metal hydroxide

B. metal sulphide

C. metal oxide

D. metal carbonate

Answer: C

Solution:

Solution:

Calcination is the process of converting an ore into its oxide by heating it strongly below its melting point either in absence or limited supply of air. It is usually done to convert metal carbonates and hydroxides to their respective oxides.

$$\underset{\text{Limestone}}{CaCO_3} \xrightarrow{\Delta} \underset{\text{Calcium Oxid}}{CaO} + CO_2\uparrow$$

$$\underset{\text{Malachite}}{CuCO_3 \cdot Cu(OH)_2} \xrightarrow{\Delta} \underset{\text{Copper Oxid}}{2CuO} + H_2O\uparrow + CO_2\uparrow$$

Question106

Which one of the following ores is known as Malachite: [Online April 19, 2014]

Options:

A. Cu_2O

B. Cu_2S

C. $CuFeS_2$

D. $Cu(OH)_2 \cdot CuCO_3$

Answer: D

Solution:

Solution:
Malachite is a green copper carbonate hydroxide mineral with the formula of $Cu(OH)_2 \cdot CuCO_3$. It was one of the first ores used to produce copper metal.

Question107

The form of iron obtained from blast furnace is: [Online April 9,2014]

Options:

A. Steel

B. Cast Iron

C. Pig Iron

D. Wrought Iron

Answer: C

Solution:

Solution:
The iron obtained from blast furnace contains about $4\%C$ and impurities $(cg.S, P, Si, Mn)$. This is known as pig iron.

Question108

The metal that cannot be obtained by electrolysis of an aqueous solution of its salts is:
[2014]

Options:

A. Ag

B. Ca

C. Cu

D. Cr

Answer: B

Solution:

Solution:
On electrolysis of aqueous solution of s -block elements, H_2 gas is discharged at cathode. At cathode:

$$H_2O + e^- \longrightarrow \frac{1}{2}H_2 + OH^-$$

Question109

Calcination is the process in which:
[Online April 9,2013]

Options:

A. ore is heated above its melting point to expel H_2O or CO_2 or SO_2

B. ore is heated below its melting point to expel volatile impurities

C. ore is heated above its melting point to remove S, As and Sb as SO_2, As_2O_3 and Sb_2O_3 respectively

D. ore is heated below its melting point to expel H_2O or CO_2

Answer: D

Solution:

Solution:
Calcination is a process of heating a substance to a high temperature but below the melting or fusion point, causing loss of moisture, reduction or oxidation and dissociation into simpler substances.

Question110

In Goldschmidt aluminothermic process which of the following reducing agents is used:
[Online April 22,2013]

Options:

A. calcium

B. coke

C. Al-powder

D. sodium

Answer: C

Solution:

Solution:
Reduction by powdered aluminium is known as Gold Schmidt aluminothermic process. This process is employed in cases where metals have very high m.p. and are to be extracted from their oxides.

Question111

Which of the oxide groups among the following cannot be reduced by carbon?
[Online May 7,2012]

Options:

A. Cu_2O, SnO_2

B. CaO, K_2O

C. PbO, Fe_2O_4

D. Fe_2O_3, ZnO

Answer: B

Solution:

Solution:
Ca and K are strong reducing agents, hence their oxides can not be reduced with carbon.

Question112

Extraction of zinc from zinc blende is achieved by
[Online May 19,2012]

Options:

A. roasting followed by self-reduction

B. electrolytic reduction

C. roasting followed by reduction with carbon

D. roasting followed by reduction with another metal

Answer: C

Solution:

Solution:
Extraction of Zn from zinc blends is achieved by roasting followed by reduction with carbon.
Reactions involved are :

$2ZnS + 3O_2 \xrightarrow{heat} 2ZnO + 2SO_2$ (Roasting)

$ZnO + C(Coke) \xrightarrow{heat} Zn + CO$ (Reduction)

Question113

The substance used as froth stabilisers in froth-floatation process is [Online May 26,2012]

Options:

A. Potassium ethyl xanthate

B. Aniline

C. Sodium cyanide

D. Copper sulphate

Answer: B

Solution:

Solution:
Froth stabilizers like cresols, aniline are added to stabilize the froth and enhance the non-wettability of the mineral particles.

Question114

Which method of purification is represented by the following equation? $Ti(s) + 2I_2(g) \xrightarrow{523K}$ $TiI_4(g) \xrightarrow{1700K} Ti(s) + 2I_2(g)$ [2012]

Options:

A. Zone refining

B. Cupellation

C. Polling

D. van Arkel

Answer: D

Solution:

Solution:
van Arkel: In this process an impure metal is converted into metal iodide which is easily decomposed to pure metal and free I_2

Question115

In the electrolysis of alumina to obtain aluminium metal, cryolite is added mainly to
[Online May 12, 2012]

Options:

A. lower the melting point of alumina

B. dissolve alumina in molten cryolite

C. remove the impurities of alumina

D. increase the electrical conductivity

Answer: A

Solution:

Solution:
Fused alumina (Al_2O_3) is a bad conductor of electricity. Therefore, cryolite (Na_3AlF_6) and fluorspar (CaF_2) are added to purified alumina which not only make alumina a good conductor of electricity but also reduce the melting point of the mixture to around $1140K$.

Question116

Which of the following factors is of no significance for roasting sulphide ores to oxides and not subjecting the sulphide ores to carbon reduction directly?
[2008]

Options:

A. Metal sulphides are thermodynamically more stable than CS_2

B. CO_2 is thermodynamically more stable than CS_2

C. Metal sulphides are less stable than the corresponding oxides

D. CO_2 is more volatile than CS_2

Answer: C

Solution:

Solution:
Note : The reduction of metal sulphides by carbon reduction process is not spontaneous because ΔG for such a process is positive. The reduction of metal oxide by carbon reduction process is spontaneous as ΔG for such a process is negative.
From this we find that on thermodynamic considerations CO_2 is more stable than CS_2 and the metal sulphides are more stable than corresponding oxides. In view of above the factor listed in choice (c) is incorrect and so is of no significance. Hence the correct answer is (c)

Question117

During the process of electrolytic refining of copper, some metals present as impurity settle as 'anode mud'. These are
[2005]

Options:

A. Fe and Ni

B. Ag and Au

C. Pb and Z n

D. Sn and Ag

Answer: B

Solution:

Solution:
During the process of electrolytic refining Ag and Au are obtained as anode mud.

Question118

Which one of the following ores is best concentrated by froth-floatation method?
[2004]

Options:

A. Galena

B. Cassiterite

C. Magnetite

D. Malachite

Answer: A

Solution:

Solution:
Note : Galena is PbS and thus purified by froth floatation method.
Froth floatation method is used to concentrate sulphide ores. This method is based on the preferential wetting properties with the frothing agent and water.

Question119

Aluminium is extracted by the electrolysis of
[2002]

Options:

A. bauxite

B. alumina

C. alumina mixed with molten cryolite

D. molten cryolite.

Answer: C

Solution:

Solution:
Pure aluminium can be obtained by electrolysis of a mixture containing alumina, crayolite and fluorspar in the ratio 20: 24: 20 . The fusion temperature of this mixture is $900^{\circ}C$ and it is a good conductor of electricity.

Question120

The metal extracted by leaching with a cyanide is [2002]

Options:

A. M g

B. Ag

C. Cu

D. Na

Answer: B

Solution:

Solution:
Silver ore forms a soluble complex with NaCN from which silver is precipitated using scrap zinc.

$$Ag_2S + 2NaCN \rightarrow \underset{\text{sodioargentocynanide (soluble)}}{Na[Ag(CN)_2]} \xrightarrow{Zn} Na_2[Zn(CN)_4] + Ag\downarrow$$
